📝 Digital SAT — Comprehensive Exam Prep

Procadamia Study Guide

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UNIT 1

📝 SAT Overview & Test Strategy

Test Structure

Section	Modules	Questions	Time
Reading & Writing	2 modules (adaptive)	54 questions	64 min (32 min each)
Math	2 modules (adaptive)	44 questions	70 min (35 min each)
Total	4 modules	98 questions	~2 hr 14 min

How Adaptive Testing Works

- **Module 1** — same difficulty for everyone
- **Module 2** — difficulty set by Module 1 performance:
  - Strong M1 → harder M2 → higher score ceiling
  - Weak M1 → easier M2 → lower score ceiling
- **Goal:** Maximize correct answers in M1 to unlock the hard M2

Scoring

Score	Percentile
1500–1600	Top 1–5%
1300–1490	Top 10–25%
1100–1290	50th–75th percentile
900–1090	25th–50th percentile

Key Strategy Rules

- ⏱️ **Pace:** RW = ~35 sec/question; Math = ~95 sec/question
- 🚩 **Flag & skip** — mark hard questions, come back within the module
- 🧮 **Use Desmos** — built-in graphing calculator on Math; use it aggressively

- ✅ **Process of elimination** — eliminate obviously wrong answers first
- ❌ **No penalty for wrong answers** — never leave a blank; always guess
- ⚠️ **Cannot go back between modules** — once you submit, that module is locked

Reading & Writing Question Types

Type	Description	~Count
Words in Context	"As used in the text, most nearly means..."	~6
Text Completion	"Which completes the text most logically/precisely?"	~8
Command of Evidence	Textual or quantitative evidence support	~8
Inferences / Claims	Best supports/weakens a claim	~6
Text Structure & Purpose	Main purpose, function of a sentence	~4
Rhetorical Synthesis	Use provided notes to complete a statement	~4
Transitions	Best transition word/phrase	~4
Boundaries / Grammar	Punctuation, sentence structure	~8

UNIT 2 Reading & Writing

📖 Words in Context

> These questions ask: *"As used in the text, what does [X] most nearly mean?"* — They test context, NOT vocabulary memorization.

The 4-Step Strategy

1. **Cover the word** — blank it out mentally
2. **Read around it** — sentence + surrounding context
3. **Predict your own word** before reading choices
4. **Match to the best answer** — closest to your prediction

> ⚠️ Trap: The test often includes the word's most common definition as a wrong answer. The word is almost always used in a more specific contextual way.

Common Trap Types

Trap	Example	Defense
Common definition trap	Picking the everyday meaning	Always re-read surrounding sentence first
Emotional tone trap	Word "feels" right for the passage	Test: does it make logical sense?
Similar-sounding trap	Confusing "venerate" with "ventilate"	Know roots

Key Vocabulary Roots

Root	Meaning	Key Word
sati-	enough / satisfy	**satiate** — fully satisfy
vener-	worship / respect	**venerate** — deeply respect

mit-/mitig-	soften / lessen	**mitigate** — reduce severity
supplant	replace / displace	**supplant** — take the place of by force
ben-	good / well	benefit, benefactor, beneficiary
temp-	time / nature	**temperament** — natural disposition

UNIT 3 Reading & Writing

✏️ Text Completion & Precision

Format: blank in a sentence — "Which choice completes the text with the *most logical and precise* word or phrase?"

Strategy: CLUE → PREDICT → MATCH

1. **Find the clue** — what does the surrounding text signal belongs in the blank?
2. **Predict** — form your own word before reading choices
3. **Match precisely** — right answer is BOTH logical AND the most specific fit

> ⚠️ Two answers may both "work" — the SAT wants the MOST precise one.

Signal Word Patterns

Pattern	Signal Words	Predict
Contrast	however, but, despite, although, yet, while	Opposite of adjacent clause
Continuation	in fact, moreover, additionally, furthermore	Same direction as stated idea
Cause-effect	because, since, therefore, thus, as a result	Logical consequence or explanation
Elaboration	for example, such as, including, specifically	More specific version of main claim

Rhetorical Synthesis Questions

- Given: student's notes (bullet points)
- Task: complete a sentence using those notes as evidence
- Strategy: read the stem → find the supporting note → eliminate choices that contradict notes or add outside info

UNIT 4 Reading & Writing

🔍 Hard Comprehension Questions

Command of Evidence — Textual

Format: "Which quote from the text best supports the claim that..."

- Find the quote that **directly proves** the specific claim
- Related ≠ right. The quote must prove the exact claim being made

Command of Evidence — Quantitative

- Includes a table, graph, or chart
- Read the claim precisely → find the exact matching data point
- Close ≠ correct — the data must precisely match the claim

Inference Questions

- ✅ Inference must be *directly supported* by the text — no outside knowledge
- ✅ "Must be true" test: if the text is true, must this answer be true?
- ❌ Eliminate extreme answers ("always," "never") unless the text explicitly says so
- ❌ Eliminate answers that go beyond what the text says

Text Structure & Purpose

Describe the **function** (why it's there), not the content (what it says).

Common functions: introduce a claim, provide evidence, present a counterargument, illustrate with an example, qualify a statement, show a contrast

Cross-Text Connection Questions

1. Summarize Passage 1 in one sentence
2. Summarize Passage 2 in one sentence
3. Identify the relationship: agree / disagree / one builds on the other
4. Answer ONLY from the text — no background knowledge

> 💡 For hard questions: go back to the passage FIRST, form your own answer, THEN read the choices. Prevents answer choices from muddying your thinking.

UNIT 5 Math

🔢 Algebra, Linear Functions & Systems

Linear Functions

- **Slope-intercept:** $y = mx + b$ (m = slope, b = y-intercept)
- **Slope:** $m = (y_2 - y_1) / (x_2 - x_1) = \Delta y/\Delta x$
- **x-intercept:** set $y = 0$, solve for x
- **Slope from table:** constant rate of change for linear functions

Systems of Equations

Type	Condition	Lines
One solution	Different slopes	Intersect at one point
No solution	Same slopes, different y-intercepts	Parallel
Infinite solutions	Same slopes AND same y-intercepts	Identical

Absolute Value Equations

- $|A| = c \rightarrow$ two cases: $A = c$ OR $A = -c$
- Combine like terms with |same expression| before splitting cases
- Example: $7|4 - x| = 49 \rightarrow |4 - x| = 7 \rightarrow x = -3$ or $x = 11$

> For "no solution" problems: set slopes equal and solve for the unknown parameter.

UNIT 6 Math

🔬 Radicals & Rational Exponents

Core Rule: a^(m/n) = $\sqrt[n]{a^m}$ — Denominator = root index; Numerator = power

Exponent Rules

Rule	Formula	Example
Product	$a^m \cdot a^n = a^{m+n}$	$x^3 \cdot x^5 = x^8$
Quotient	$a^m / a^n = a^{m-n}$	$x^7 / x^3 = x^4$
Power of a Power	$(a^m)^n = a^{mn}$	$(x^3)^4 = x^{12}$
Negative exponent	$a^{-n} = 1/a^n$	$x^{-2} = 1/x^2$
Zero exponent	$a^0 = 1$ ($a \neq 0$)	$7^0 = 1$
Fractional exponent	a^(m/n) = $\sqrt[n]{a^m}$	8^(2/3) = 4

Quick Conversions

Expression	Simplified
8^(1/3)	$\sqrt[3]{8} = 2$
8^(2/3)	$(\sqrt[3]{8})^2 = 4$
27^(1/3)	$\sqrt[3]{27} = 3$
16^(3/4)	$(\sqrt[4]{16})^3 = 2^3 = 8$
32^(2/5)	$(\sqrt[5]{32})^2 = 2^2 = 4$

UNIT 7 Math

🧮 Polynomials & Factoring

Factor Theorem

$(x - c)$ is a factor of $f(x)$ if and only if **$f(c) = 0$**. To test a factor: substitute and check result.

Key Factoring Patterns

Pattern	Formula	Example
Difference of Squares	$a^2 - b^2 = (a+b)(a-b)$	$x^2 - 9 = (x+3)(x-3)$
Perfect Square	$a^2 \pm 2ab + b^2 = (a \pm b)^2$	$x^2 + 6x + 9 = (x+3)^2$
Trinomial (a=1)	$(x+p)(x+q)$: $p+q=b$, $pq=c$	$x^2 - 2x - 15 = (x-5)(x+3)$
Trinomial (a≠1)	AC method or grouping	$3x^2+20x-63 = (x+9)(3x-7)$

Rational Expressions

Factor numerator and denominator completely, then cancel common factors.

Example: $(x^2-2x-15)/(x^2-9) = (x-5)(x+3) / (x+3)(x-3) =$ **$(x-5)/(x-3)$**

Division Identity Trick

If $P(x)/(tx-a) = Q(x) + R/(tx-a)$, then $P(x) = Q(x)\cdot(tx-a) + R$. Multiply out and match coefficients to find unknown constants.

UNIT 8 Math

⚙️ Functions & Composite Functions

Composite Functions

f(g(x)) — Work inside → out. Evaluate g(x) first, then plug into f.

f(g(a)) ≠ g(f(a)) — order matters.

Key Function Concepts

Concept	Meaning	SAT Application
f(g(x))	Apply g first, then f	Substitute g(x) for every x in f
f(a) = 0	a is a zero/root	Graph crosses x-axis at (a, 0); (x−a) is a factor
Rate of change	Slope of linear function	For f(x) = mx + b, rate = m

Graph Transformations

Transformation	Effect
f(x) + k	Shift UP k units
f(x) − k	Shift DOWN k units
f(x + h)	Shift LEFT h units
f(x − h)	Shift RIGHT h units
−f(x)	Reflect over x-axis
f(−x)	Reflect over y-axis

💡 For f(x + h) vs f(x − h): the sign inside shifts the OPPOSITE direction. f(x+3) shifts LEFT. f(x−3) shifts RIGHT.